Stock Price Prediction of Top Technology Companies Using Market Data and News Sentiment Analysis

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Abstract

This project predicts the stock prices of the six biggest technology companies, Amazon, Microsoft, NVIDIA, Facebook, Google, and Apple, using stock price data retrieved via the Tiingo API and financial news data from the Financial Modeling Prep (FMP) API. By analyzing real-time and historical stock price data alongside relevant financial news, the study applies machine learning algorithms to identify trends and forecasts over the next 15 trading days after March 4th, 2025. Overall, we find the LSTM model performs the most consistently for forecasting across all companies given different data cleaning and model tuning. These findings could offer valuable insights for investors and contribute to advancements in financial data science. [1]

1 Introduction

Recent studies have shown a growing use of data science and forecasting techniques to predict stock market behavior. [2] Stock price prediction is crucial for investors, analysts, and policymakers, as it is influenced by numerous factors such as economic data, company performance, and market sentiment. Advances in big data and machine learning have enhanced the ability to analyze large datasets and identify patterns, making more accurate forecasts possible. This project aims to leverage these advancements to predict stock prices for six leading technology companies—Amazon, Microsoft, NVIDIA, Facebook, Google, and Apple.

However, existing research often relies on recent stock data, missing long-term trends, seasonality, and AI-driven market shifts. Advances in generative AI, automated trading, and algorithmic decision-making have added complexity, making traditional models less effective. For example, NVIDIA, a leader in AI-driven computing, saw a stock downturn following shifts in AI demand and regulatory concerns. This study aims to improve stock price forecasting by integrating historical stock data with sentiment analysis of financial news, capturing both market trends and external influences.

These companies were chosen for their significant influence on the global economy and the availability of extensive stock price data. Stock price data will be retrieved from the Tiingo API, covering each company's prices from its Initial Public Offering (IPO) date onward, while financial news and all other data will be sourced from the FMP API, with news coverage starting from 2011 onwards, except for Facebook (2019) and NVIDIA (2017). By integrating stock price trends with market sentiment analysis and applying machine learning algorithms, this study seeks to improve forecasting accuracy and provide deeper insights into stock behavior. The findings from this research could support better decision-making for investors and contribute to advancements in financial data science.

2 Materials: Data Gathering & Preprocessing

A visual breakdown of the entire data collection and modeling process is as follows:

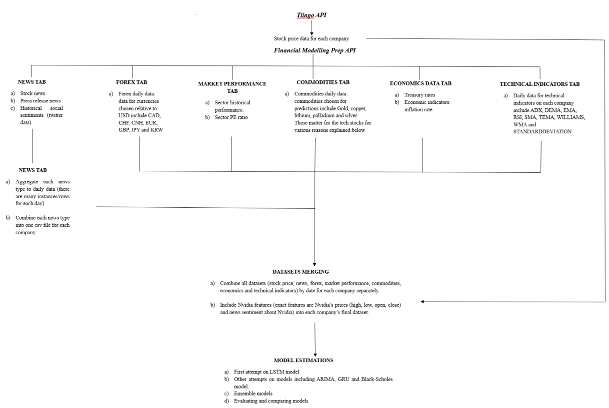

Figure 1: Data Collection and Modelling Breakdown

2.1 Stock Price Data

The Tiingo API was used to retrieve historical stock price data for major technology companies, including Amazon (AMZN), Microsoft (MSFT), Meta (META), Nvidia (NVDA), Google (GOOGL), and Apple (AAPL). The data spans from each company's Initial Public Offering (IPO) date to the current date. The Tiingo API is a comprehensive financial data service that provides users with access to a wide range of stock market data, including key parameters such as open, high, low, close, adjusted close prices, and trading volume for a specific stock symbol. An IPO is the process by which a private company offers shares of its stock to the public for the first time, marking the company's transition from being privately owned to publicly traded on a stock exchange. The IPO allows the company to raise capital from public investors, which can be used for expansion, debt reduction, or other corporate purposes. By retrieving stock price data from the IPO date to the present, we can track the performance of a company's stock over time, which is crucial for building forecasting models. This historical stock price data provides critical insights into market behavior, trends, and volatility. It helps us understand past performance, spot emerging trends, and measure market volatility - all of which are essential factors for stock price prediction. The Tiingo API facilitates seamless access to this data, enabling us to gather comprehensive datasets and develop robust forecasting models that can better predict future stock movements.

2.2 News Data

Deep learning is an effective tool for analyzing unstructured data, and it is increasingly applied to stock market forecasting, but often neglects the impact of mass media on stock prices and investor behavior. [3] uses an hybrid model combining Word2Vec and Long Short-Term Memory (LSTM) to predict stock price movements using financial data and news headlines. The following paper [4] highlights the use of sentiment analysis of stock-related tweets using NLP technology that improves prediction efficiency. A CNN model was developed using sentiment data and stock price data to predict and forecast stock movements. As cited in [5] Valence Aware Dictionary and sEntiment Reasoner (VADER), a sentiment analysis tool is being used to determine whether a particular tweet has a positive, neutral or negative sentiment.

To extensively capture the impact of mass media, we pull stock news, press release news and twitter social sentiments from the FMP API for each tech company. The stock news dataset covers relevant financial news from 2011 onwards for most companies, except for Facebook (2019 onwards) and NVIDIA (2017 onwards). The press release news data covers a similar range for the companies.

To investigate the impact of social sentiment on stock movement, we collected data on Twitter (X) posts related to the tech companies included in our model. Using the FMP API, we retrieved tweets mentioning these stocks, along with engagement metrics and sentiment scores. These sentiment scores help assess the influence of social media discussions on stock price fluctuations. Prior research suggests that social media activity, including tweets, can impact stock prices [6]. Given this, we aimed to incorporate social media sentiment—often less vetted than traditional news sources—into our analysis to explore its role in stock price movements. However, a key limitation of the FMP API is that it only provides social sentiment data for the past three months, restricting our dataset to information from December 2024 to March 4th, 2025.

After retrieving all news-related data, a text cleaning function is created to remove any urls, irrelevant characters and extra spaces in the headlines and article summaries of the news content. Furthermore, all (%) signs are converted to "percentages" and abbreviations such as "M" or "B" are converted to "million" and "billion." This is done to better process the financial news text for sentiment score calculation, although VADER does not require it to be done.

We apply VADER, to assess the sentiment of the headlines and article summaries. VADER, which is pre-trained, is effective in processing words and abbreviations, making it an easy tool for determining article sentiment. It assigns a sentiment score to each word, ranging from -1 (highly negative) to 1 (highly positive). The overall sentiment of the article is then determined by averaging the scores of all the words in the headline and summary. Additionally, VADER can analyze emoticons, which could be valuable when including social media posts or broader news articles from the FMP API, offering a more comprehensive understanding of public sentiment towards the companies on a given day.

Once the sentiment score is calculated, it is checked against a threshold and placed in a resulting category. When the score is less than -0.5, "strongly_negative" is assigned, greater than 0.5, "strongly_positive", between 0.05 and 0.5, "weakly_positive", between -0.05 and -0.5, "weakly_negative" else "neutral".

This sentiment score calculation is applied twice. First, on the raw news data which have many news article instances for each day for each company. Another function is created to take the mean of these sentiment scores by day. The thresholds are reapplied to the sentiment score for each day and classified accordingly. Finally, the stock news, press release news and twitter social sentiment are combined into one excel file for each company.

2.3 Forex Data

[7] examines the conditional correlation between exchange rates and stock prices. Covering five exchange rates: US Dollars (USD) vs. Great British Pounds (GBP), Canadian Dollars (CAD), Deutsche Mark (DEM), Euro (EUR) and Japanese Yen (JPY), his findings indicate that the correlation between exchange rates and stock prices is time-varying and currency-dependent. His model suggests that the correlation is determined by the relative sensitivity of two countries' stock prices to a common risk factor. When a positive shock occurs, the country with higher sensitivity attracts more investment, leading to currency appreciation. Hence, we pulled data covering the above-mentioned currencies all relative to the USD with the exception of DEM.

We also pulled offshore Chinese Yuan (CNH), which indicates China's economic health, affecting crucial supply chains for tech companies, the Korean Won (KRW), since South Korea is a semiconductor powerhouse, affecting Nvidia and tech supply chains, and the Swiss Franc (CHF), a risk aversion currency that can impact investor behavior in tech stocks. The data range on these currencies covered from 5th March, 2020 to 3rd March, 2025 as allowed by the FMP API.

2.4 Market Performance Data

The FMP API provided valuable information on market performance for the technology sector, as well as other industries, such as defense and financial services, that could influence the performance of tech stocks. Using the Sector Historical API, we retrieved historical data on sector performance changes from December 12, 1980, to March 4, 2025. Rather than isolating only the technology sector, we retained data for all sectors to examine potential cross-sector influences on technology stock prices.

Additionally, we obtained the Sector Price-to-Earnings (PE) Ratio from the Market Performance API. The PE ratio is a key metric that indicates how expensive a stock is relative to its earnings. A high PE ratio may suggest that a stock is overvalued, reflecting investor expectations of higher future earnings, while a low PE ratio may indicate undervaluation ([8], [9]). Since market participants often rely on the PE ratio when making investment decisions, fluctuations in this metric can influence stock prices, making it a valuable factor to consider in our analysis.

2.5 Commodities Data

Gold, alongside silver is widely viewed as a safe-haven asset. In times of market stress or economic uncertainty, investors often move money into gold and/or silver, reducing their exposure to riskier assets like tech stocks ([10], [11]). Also, several commodities including silver [11], lithium [12], palladium [13] and copper [14] are used in the tech industry in circuit boards, chips and many more applications. Their availability to these tech companies such as Apple tends to influence their operations and thus investor behavior. Data gathered on commodities ranges from January 1980 to March 3rd, 2025.

2.6 Economics Data

[15] discover a negative relationship between inflation rate and stock returns for US aggregated stock indices in different sectors including financials and technology. [15] also discover a similar effect in other G7 countries and argued that with lower inflation rates, traders buy more stocks or go long on their stocks. On the other hand, increased inflation rates have a mixed effect as some rational traders delay shorting their stocks to gather more intel whereas others do short their stocks. The net effect, however, from their study is a negative relationship. Hence, we gathered daily inflation data ranging from January 2003 to 3rd March, 2025 on the US economy as it is crucial to investor behaviour.

We also gathered data on treasury rates ranging from January 1990 to 3rd March, 2025. Treasury rates form the base for which banks can set their interest rates. Increases in China's official interest rate affects the NYSE-listed Chinese stocks negatively and U.S based investors take advantage of this effect and buy more Chinese stocks and vice versa when official interest rates in China decreases [16]. A similar analogy can be made for U.S. stocks as was evident during the 2008 financial crisis [16].

2.7 Technical Indicators Data

The technical indicators which are retrieved from the Financial Modeling Prep (FMP) API are mathematical calculations based on stock price and trading volume data, designed to help forecast stock price trends and market behavior. These indicators, such as ADX (Average Directional Index), DEMA (Double Exponential Moving Average), EMA (Exponential Moving Average), RSI (Relative Strength Index), SMA (Simple Moving Average), and others, provide insights into market momentum, trend strength, volatility, and potential price reversals. By analyzing these indicators in combination with historical price data, we can better predict future market movements. The data for each stock was retrieved in 4-year spans due to limitations in the API, we were unable to extract all data in a single request.

Stocks such as Amazon, Apple, Google, Microsoft, Meta, and Nvidia were analyzed from their respective IPO dates up to the current date. Among these, indicators such as RSI, ADX, and EMA are particularly important in stock forecasting, as they help identify trends, momentum, and market conditions such as overbought or oversold situations. The data for these indicators were retrieved by making API requests for each stock in time intervals, and the resulting data were stored in CSV files for analysis. The combination of technical indicators and historical price data is then used to create more accurate forecasting models for predicting stock price movements.

2.8 Data Merging & Preprocessing

All the datasets mentioned above are joined together for each company. Furthermore, stock price features and news sentiment from Nvidia are added as additional features to other tech company's datasets. This is done as companies such as Google and Apple rely heavily on Nvidia's GPU chips and other products. When Nvidia stock prices suddenly plummet or skyrocket, they are bound to influence other tech stocks directly. After including these features, the datasets for each tech company are complete and ready for cleaning.

Data preprocessing will involve handling missing values, filtering noise, and ensuring consistency. Consistency with the data is key as the features do not have the same data periods. Particularly, majority of the features do not go as far back as the IPO dates for each company like the stock prices do. Thus, for each company, the start of the data period would be decided based on the range in which majority of the features are available.

A range of data cleaning techniques was applied to the datasets that were created. Missing values were addressed using methods such as mode, monthly averages, rolling means, and linear interpolation, depending on the characteristics of each variable. The FMP API did not provide sufficiently complete data for the social sentiment columns; only three months of data were available, resulting in significant gaps. Due to the extent of missing data, the social sentiment columns were excluded from the analysis.

For the stock news and press release sentiment data available through the API, sentiment classification was performed using the GPT-4o large language model (LLM). The model categorized sentiment on a five-point scale: very negative, slightly negative, neutral, slightly positive, and very positive. Due to limited access to the LLM, not all available news entries could be evaluated. For missing sentiment values, the mode for each respective month was used as a proxy to approximate the overall sentiment during that period.

3 Methods

To model and predict stock prices based on the combined data from stock prices, financial news sentiment, and other features from FMP, this study will explore various machine learning and deep learning algorithms. Among the techniques considered are LSTM, AutoRegressive Integrated Moving Average (ARIMA), Lasso regression, and Bidirectional Long Short-Term Memory (BiLSTM). Additionally, we may explore hybrid models that combine the strengths of different algorithms, such as integrating LSTM with ARIMA for time series forecasting or using CNN for sentiment analysis in conjunction with LSTM for stock price prediction.

LSTM (Long Short-Term Memory) networks are a specialized type of Recurrent Neural Network (RNN) designed to handle long-term dependencies in sequential data. Unlike traditional RNNs, which struggle with the vanishing gradient problem when learning from long sequences, LSTMs can effectively retain information over extended periods. This makes them well-suited for tasks that require an understanding of temporal relationships, such as natural language processing, speech recognition, and time-series forecasting. LSTMs operate by maintaining a cell state, which functions as memory, and using three gates—forget, input, and output—to control the flow of information. The forget gate determines what information should be discarded from the previous time step, the input gate decides what new information should be added to the memory, and the output gate controls what part of the memory should be used for generating the current output. These mechanisms allow LSTMs to remember relevant information while forgetting unnecessary details, addressing the challenges faced by traditional RNNs. This is particularly useful for tasks like speech recognition, time-series forecasting, and natural language processing, where long-term context is crucial.

When applied to stock price forecasting, LSTM networks excel due to their ability to capture long-term dependencies in time-series data, such as stock prices, trading volumes, and other financial indicators. Historical stock data is often preprocessed (normalized and divided into time windows or sequences) to predict future stock prices. The LSTM model is trained on these sequences to learn the temporal patterns and dependencies within the data, allowing it to predict future stock prices based on the most recent data. The performance of the model is typically evaluated using metrics like Mean Squared Error (MSE), and predictions can be made for several time steps ahead. LSTMs' ability to handle complex, non-linear relationships in the stock market and adapt to changing market conditions makes them highly effective for forecasting.

To further improve stock price predictions, news sentiment data can be integrated into the LSTM model. News articles, financial reports, and social media content often contain valuable insights about market sentiment that directly affect stock prices. By using NLP techniques, news data can be processed to extract sentiment indicators (e.g., positive, negative, or neutral sentiment). The sentiment scores can be used as additional input features alongside stock price data. In practice, the news sentiment data is typically processed through sentiment analysis models that convert raw text into numerical sentiment scores. These scores, representing the emotional tone of the news (positive or negative), are then fed into the LSTM model along with the historical stock data. The LSTM network can learn to combine both historical stock prices and news sentiment to make more informed predictions, improving forecasting accuracy by taking into account not just past price trends but also market sentiment. This multi-modal approach allows the model to capture both the historical behavior of stock prices and the real-time effects of external factors such as economic announcements, company news, or global events, ultimately enhancing the ability to predict stock price movements.

ARIMA, and its sister models like SARIMAX, is primarily used for time-series forecasting based on historical data, but it can be extended to include news data for more accurate predictions, particularly in domains like stock price forecasting. To incorporate news data into an ARIMA model, sentiment analysis is typically applied to extract meaningful information from news articles, financial reports, or social media posts. The sentiment analysis process evaluates the emotional tone of the news (positive, negative, or neutral), which can then be quantified into sentiment scores or indicators.

These sentiment scores are treated as exogenous variables (external regressors) in the ARIMA model. By including sentiment scores along with the time-series data, the model can adjust its predictions based on the real-time impact of news events. For instance, if positive news about a company or the economy is detected, the sentiment score will be higher, and this can influence the ARIMA model to predict a more favorable future stock price. Conversely, negative news would lead to a lower sentiment score and could adjust predictions downwards.

Thus, while the ARIMA model alone would rely on historical price data to forecast future values, combining it with news sentiment data provides a more dynamic approach, capturing both the underlying historical patterns and the immediate market reactions to external events. This integration enhances the model's ability to reflect the influence of current events and sentiments, offering more accurate and timely predictions, especially in fast-moving markets where news can significantly impact prices in the short term.

BiLSTM networks are an extension of standard Long Short-Term Memory (LSTM) networks. While traditional LSTMs process input sequences in a single forward direction (from past to future), BiLSTM networks utilize two LSTMs—one processing the sequence in the forward direction and another in the reverse direction. This allows BiLSTMs to capture both past and future dependencies within sequential data, making them highly effective for time-series forecasting and natural language processing tasks.

Unlike standard LSTMs, which rely only on past information when making predictions, BiLSTMs leverage information from both past and future contexts. This is particularly advantageous in applications where a complete understanding of temporal relationships is necessary. The dual processing mechanism helps BiLSTMs retain more context, leading to improved accuracy in tasks that require a comprehensive view of sequential patterns.

In the context of stock price forecasting, BiLSTMs are advantageous because stock prices often exhibit complex dependencies where future prices may be influenced by both past trends and external factors such as market news and investor sentiment. By processing historical stock data in both forward and backward directions, BiLSTM networks can uncover intricate patterns that standard LSTMs might overlook.

Similar to LSTMs, BiLSTM models require data preprocessing, including normalization and sequence generation. They are typically trained using optimization techniques like Adam or RMSprop and evaluated using performance metrics such as Mean Squared Error (MSE) and Mean Absolute Error (MAE). Additionally, BiLSTM models can be combined with news sentiment analysis to incorporate external market factors, further enhancing predictive capabilities.

By capturing bidirectional dependencies, BiLSTMs provide a more comprehensive understanding of stock price movements, making them particularly effective in volatile markets where sudden shifts in trends may occur due to external news or investor behavior.

LASSO(Least Absolute Shrinkage and Selection Operator) is a linear regression technique that performs both feature selection and regularization, making it highly effective for stock price prediction. Stock market data includes numerous variables such as historical prices, trading volume, financial indicators, and even news sentiment, many of which may be redundant or irrelevant. LASSO addresses this by applying L1 regularization, which forces some regression coefficients to exactly zero, effectively removing less significant features and reducing the risk of overfitting. This makes the model more interpretable and robust, especially in high-dimensional datasets. In stock price prediction, LASSO helps identify the most important predictors while discarding noise, improving generalization to unseen data. It is particularly useful for handling multicollinearity, as it selects only one of the highly correlated features rather than distributing weight across them. The process involves collecting and preprocessing stock market data, selecting an optimal regularization parameter through cross-validation, and training the model to predict future stock prices based on the most relevant factors. Model performance is typically evaluated using metrics like Mean Squared Error (MSE) or Mean Absolute Error (MAE), and LASSO is often compared with other predictive models such as Ridge regression, LSTM networks, or ARIMA to determine its effectiveness.

The final selection of models will be based on performance metrics to determine the most efficient and reliable approach to stock price prediction.

4 Results

In this analysis, we use several models to forecast stock prices for Microsoft, Amazon, Nvidia, Google, Apple, and Meta. The models are LSTM, BiLSTM, ARIMA, Random Forest, and Lasso regression. The forecasted prices for each model and company are displayed, with a comparison of the model performances based on the Root Mean Squared Error (RMSE). This comparison helps determine which model offers the most accurate predictions for stock prices across these major tech companies, providing insights into the effectiveness of different forecasting techniques in the financial sector.

4.1 AMAZON

4.1.1 LSTM

LSTM model was used to predict Amazon's stock closing prices using historical data. It preprocesses the data by extracting date-related features and normalizing the closing prices. The data is split into training and testing sets, and sequences of 60 time steps are created for the LSTM input. The model consists of two LSTM layers with dropout for regularization, followed by Dense layers for prediction. After training, the model's performance is evaluated using RMSE, and actual vs. predicted prices are plotted. Finally, the model is used to forecast stock prices for the next 15 days, which are also plotted. In figure 2, the corresponding forecasted values are displayed below and in a tabular format in the Appendix.

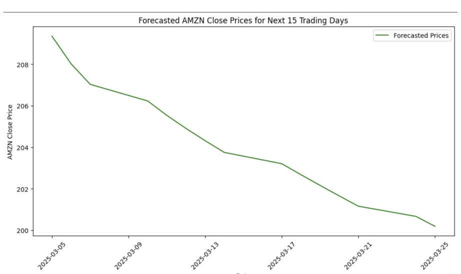

Figure 2: LSTM forecasted amazon stock price plot

4.1.2 BiLSTM

Bidirectional LSTM model was used to predict Amazon's stock closing prices. The data is preprocessed by extracting date-related features and normalizing the closing prices using MinMaxScaler. It is then split into training and testing sets, with sequences of 60 time steps created for LSTM input. The model, consisting of Bidirectional LSTM layers and dropout for regularization, is trained to predict future stock prices. After training, the model's performance is evaluated using RMSE, and actual vs. predicted prices are plotted. Additionally, the model forecasts Amazon's stock prices for the next 15 trading days, which are also visualized in Figure 3.

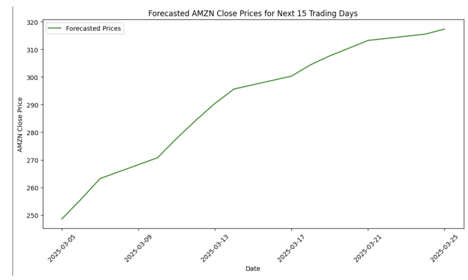

Figure 3: BiLSTM forecasted amazon stock price plot

4.1.3 Lasso

Lasso regression was used to predict Amazon's stock closing prices. It preprocesses the data by standardizing the features and creating sequences of 60 time steps for input. The data is split into training and testing sets, and the sequences are flattened for Lasso model training. After fitting the model, it evaluates performance using RMSE and plots actual vs. predicted prices. The model is then used to forecast the next 15 trading days' stock prices, which are plotted as well in Figure 4.

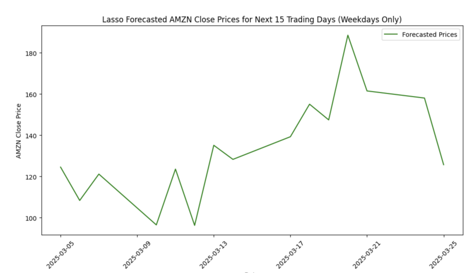

Figure 4: Lasso forecasted amazon stock price plot

4.1.4 Random Forest

Random Forest Regressor model was used to predict Amazon's stock closing prices. The data is preprocessed by standardizing the features and creating sequences of 60 time steps. It splits the data into training and testing sets, trains the model, and evaluates performance using RMSE. Actual vs. predicted prices are plotted for the test set. The model is then used to forecast stock prices for the next 15 trading days, and the forecasted prices are plotted Figure 5.

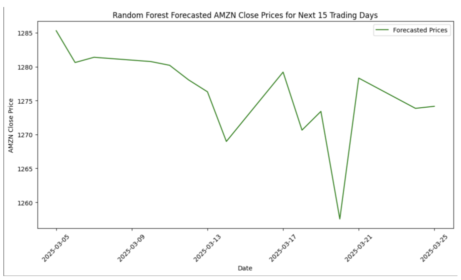

Figure 5: Random Forest forecasted amazon stock price plot

4.1.5 ARIMA

ARIMA model was used to forecast Amazon's stock closing prices. It begins by preprocessing the data, including converting the 'date' column to a Unix timestamp and splitting the data into training and testing sets based on a quantile split. The training data is then scaled using a MinMaxScaler. The ARIMA model, with order (8, 2, 9), is fitted to the scaled training data, and predictions are made for the test set. The predicted values are then inverse-transformed back to the original scale, and the RMSE is computed to assess the model's accuracy. A plot compares the actual vs. predicted prices for the test set. Additionally, the model is used to forecast the next 15 trading days, and these forecasted prices are plotted alongside the actual data, offering insights into future price trends in Figure 6.

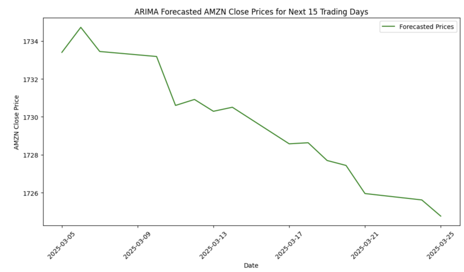

Figure 6: ARIMA forecasted amazon stock price plot

4.1.6 Model Comparison

Figure 7 compares the performance of five different models—LSTM, BiLSTM, Lasso Regression, Random Forest, and ARIMA—on forecasting stock prices for companies like Microsoft, Amazon, Nvidia, Google, and Meta to the actual stock prices for the same 15 trading days. The models are evaluated using multiple metrics: Pearson correlation, Spearman correlation, RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and MAPE (Mean Absolute Percentage Error).

From the table, the LSTM model performs the best, with the highest Pearson (0.2177) and Spearman (0.1429) correlations, and a relatively low RMSE (7.36), MAE (6.84), and MAPE (3.48). In contrast, the BiLSTM model, while it has slightly better correlations than other models, suffers from a very high RMSE (41.91) and MAE (40.56), indicating poor predictive accuracy.

Lasso Regression and Random Forest also show subpar performance, with Lasso Regression having the highest RMSE and MAE values, and Random Forest showing a significant RMSE (1077.64) and MAPE (543.73), indicating considerable errors in forecasts.

ARIMA, while often a reliable model for time series data, also shows poor performance here with the highest RMSE (1531.43), MAE (1531.42), and MAPE (772.71), indicating that it is not suitable for this particular stock price forecasting task.

In conclusion, LSTM stands out as the best model, outperforming others across multiple evaluation metrics, making it the preferred choice for stock price prediction tasks in this analysis.

	Model	Pearson Correlation	Spearman Correlation	RMSE	MAE	MAPE
0	LSTM	0.217733	0.142857	7.361995	6.842141	3.475202
1	BiLSTM	-0.251346	-0.142857	41.909390	40.557629	20.533656
2	Lasso Regression	-0.241101	-0.328571	69.281401	64.307179	32.333305
3	Random Forest	0.349871	0.360714	1077.636755	1077.617631	543.725519
4	ARIMA	0.045094	0.132143	1531.431446	1531.422710	772.712212

Figure 7: Performance of all the models for Forecasted Data

4.2 MICROSOFT

4.2.1 LSTM

Microsoft's stock prices was predicted using an LSTM model. It begins by preprocessing the data, extracting date-related features, and scaling the stock prices. The data is split into training and test sets based on a quantile of the Unix timestamp. Sequences of 60 time steps are created for the LSTM, which is trained to predict future stock prices. The model's performance is evaluated using RMSE, and actual vs. predicted prices are plotted. Additionally, the model forecasts the next 15 days of stock prices, and the results are visualized in Figure 8. The LSTM model is structured with two layers, dropout for regularization, and is trained using the Adam optimizer with a mean squared error loss function.

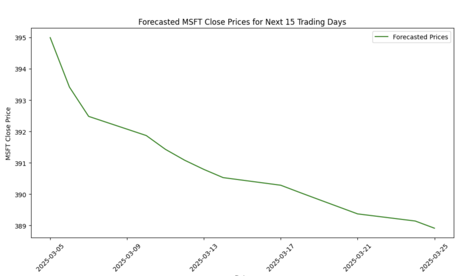

Figure 8: LSTM forecasted microsoft stock price plot

4.2.2 BiLSTM

Bidirectional LSTM model was used to predict Microsoft's stock prices. It starts by preprocessing the data, extracting date-related features, and scaling the stock prices. The data is split into training and test sets based on a quantile of the Unix timestamp. Sequences of 90 time steps are created for the LSTM, which is trained to predict future stock prices. The model's performance is evaluated using RMSE, and actual vs. predicted prices are plotted. A forecast for the next 15 days is generated, and the results are visualized Figure 9. The Bidirectional LSTM enhances the model by processing data in both forward and backward directions, which captures more context from the time series.

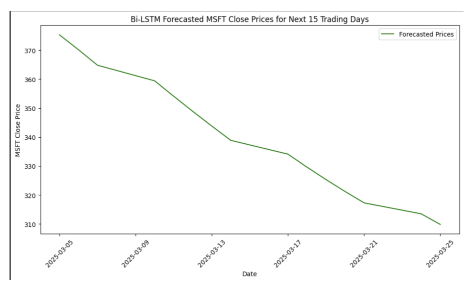

Figure 9: BiLSTM forecasted microsoft stock price plot

4.2.3 Lasso

Lasso regression was used to predict Microsoft's stock prices by processing time-series data. After dropping missing values and standardizing the features, the data is split into sequences for the model. A train-test split is done, and the sequences are flattened to fit the Lasso model. The model is trained on the training set, and its performance is evaluated using RMSE. Actual vs. predicted values are plotted to visualize the fit. For future predictions, the last known sequence is used iteratively to forecast the next 15 trading days, excluding weekends, and these forecasted prices are plotted in Figure 10. The use of Lasso regularization helps prevent overfitting, especially with a large number of features.

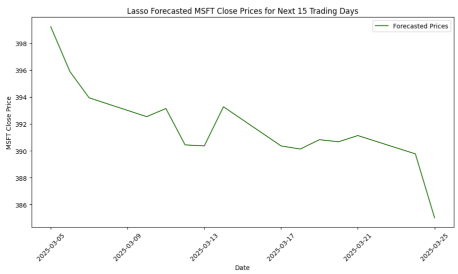

Figure 10: Lasso forecasted microsoft stock price plot

4.2.4 Random Forest

Random Forest model was used to predict Microsoft stock prices using lag features and moving averages. First, it creates lagged features and rolling averages (5-day and 10-day) for the stock price. After standardizing the features, the data is split into sequences for the model, and a train-test split is applied. The Random Forest model is trained on the flattened sequences, and its performance is evaluated with RMSE. Predictions are made for the test set, and actual vs. predicted prices are plotted. For forecasting the next 15 trading days, the last known sequence is iteratively used to predict future values, excluding weekends, and these forecasted prices are plotted in Figure 11. The use of Random Forest helps capture non-linear relationships in the time-series data.

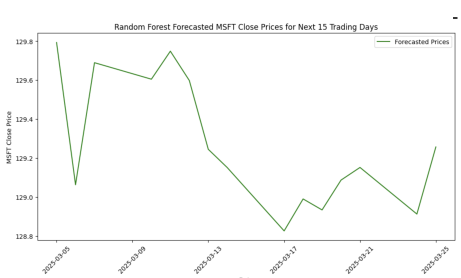

Figure 11: Random Forest forecasted microsoft stock price plot

4.2.5 ARIMA

ARIMA model was used to forecast Microsoft stock prices. After preparing the data by removing columns related to NVDA and scaling the MSFT close prices, it splits the dataset into training and test sets based on the date. The ARIMA model is then trained on the scaled training data with an order of (8, 2, 9). Predictions are made on the test set, and RMSE is calculated to evaluate model performance. Actual and predicted values are plotted for comparison. The code also forecasts the next 15 trading days, considering only weekdays, and plots the forecasted values in Figure 12. The ARIMA model captures temporal dependencies in the stock price data and provides a forecast for future prices.

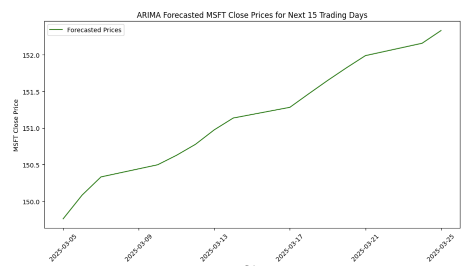

Figure 12: ARIMA forecasted microsoft stock price plot

4.2.6 Model Comparison

Figure 13 compares the performance of five models—LSTM, BiLSTM, Lasso, Random Forest, and ARIMA—on a stock price forecasting task. The models are evaluated using Pearson correlation, Spearman correlation, RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and MAPE (Mean Absolute Percentage Error).

From the table, LSTM performs better than the other models in this task. Despite having weak correlations, LSTM achieves a relatively low RMSE (259.40) and MAE (259.32), indicating that it is able to capture trends in stock prices better than models like BiLSTM, Random Forest, and ARIMA. While Lasso Regression shows strong correlations and lower error metrics, LSTM outperforms it in overall prediction capability for stock price forecasting.

In conclusion, LSTM is the best-performing model for this task, balancing reasonable accuracy and the ability to predict stock prices effectively despite the complexity of the task.

	Model	Pearson Correlation	Spearman Correlation	RMSE	MAE	MAPE
0	LSTM	-0.049535	-0.057143	259.396330	259.317663	200.604515
1	BiLSTM	0.082550	0.014286	52.528512	48.190387	14.562996
2	Lasso	0.326509	0.246429	6.885725	5.377547	1.375676
3	Random Forest	-0.049535	-0.057143	259.396330	259.317663	200.604515
4	ARIMA	-0.071868	-0.014286	237.549176	237.461255	157.135921

Figure 13: Performance of all the models for Forecasted Data

4.3 META

To enhance the model's ability to capture temporal dependencies, lagged features were generated for the Meta adjusted closing stock price. Lags were created to reflect the influence of past values on future price movements. In addition to lagging the target variable (`META_adjClose`) by 1 to 7 days, rolling statistics such as 5-day rolling means and standard deviations were computed to account for short-term trends and volatility.

Lagged features were also created for selected external variables believed to influence Meta's stock price. These included:

- `inflationRate` (lagged by up to 60 days to capture long-term macroeconomic effects),
- `Gold_price` and `Silver_price` (lagged up to 7 days to reflect short-term commodity influences),
- `META_stock_news_sentiment` (lagged up to 5 days to reflect immediate sentiment effects), and
- `NVDA_stock_news_sentiment` (lagged up to 7 days as a proxy for sector-related sentiment).

Sentiment values, originally categorical (e.g., very positive, neutral, ect.), were manually encoded on a numeric scale from -1 (very negative) to 1 (very positive). This conversion was applied across all lagged sentiment columns to prepare them for numerical modeling.

The resulting feature set was stored in a new DataFrame for modeling purposes. After preprocessing and alignment with the original time index, the models were trained on the entire historical dataset. Forecasts were then generated for the 15 trading days following March 4^{th}, 2025, using the most recent known values and recursively updating the lag structure with each prediction.

4.3.1 LSTM

A simple LSTM model was found to perform best on the Meta stock data. The current model architecture consists of a single LSTM layer with 50 units, using one time step and two features per time step, derived from lagged values. A Dense output layer with one unit is used to generate a single price prediction per day. The model is compiled with the Adam optimizer and mean squared error (MSE) as the loss function.

Alternative configurations were tested, including additional LSTM layers, various dropout rates, and the use of batch normalization. These more complex models generally resulted in reduced performance. Forecast outputs tended to become either overly exaggerated—showing sharp slopes and turning points—or excessively flat, failing to capture directional changes accurately.

This behavior is likely due to the erratic and noisy nature of stock price movements, which can make it difficult for deeper models to distinguish signal from noise. Larger time steps were also explored but exhibited similar issues. Overall, a simpler LSTM architecture proved more effective for this dataset. Figure 14 shows the performance of the LSTM model compared with the true prices for the true prices for that time period.

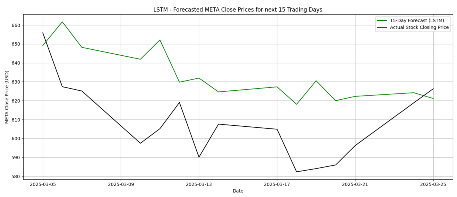

Figure 14: LSTM Prediction of closing stock price for META

4.3.2 Bi-LSTM

Due to the noisy nature of the data and consistent with the approach taken for the standard LSTM model, the Bidirectional LSTM (Bi-LSTM) architecture was kept simple. The model consists of a single Bidirectional LSTM layer with 32 units and an input shape comprising one time step and multiple features. A Dense output layer with one unit was used to predict a single stock price per time step. The model was compiled with the Adam optimizer and mean squared error (MSE) as the loss function. Figure 15 shows the performance of this model.

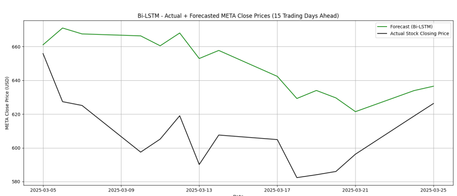

Figure 15: Bi-LSTM Prediction of closing stock price for META

4.3.3 Lasso

The Meta Lasso regression model was first tuned using K-Fold cross-validation with shuffling to identify the optimal value of the regularization parameter, alpha. Once the best alpha was selected, a final model was trained and evaluated using TimeSeriesSplit, a time-aware cross-validation method that preserves the temporal ordering of the data. This ensured the model was validated in a realistic forecasting scenario by only using past data to predict future values.

The model used lagged versions of the target variable and selected external features as input. After training, the model was used to generate a 15-day ahead forecast of Meta's adjusted closing stock price. Forecasts were generated recursively: each new predicted value was added back into the lagged feature structure to predict the next time step. External lagged features were rolled forward as well, with the most recent values held constant or reused in the absence of future data.

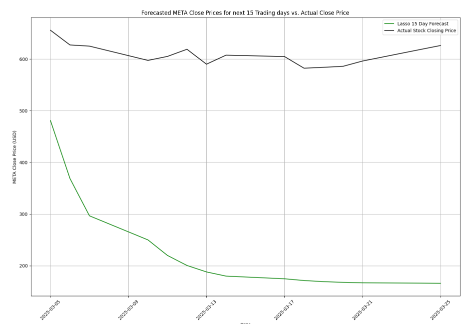

Figure 16: Lassp prediction of closing stock price for META

4.3.4 SARIMA

To try and account for the seasonality of stock prices, the SARIMAX model, a version of the ARIMA model, was used. This model can incorporate both autoregressive components and exogenous variables in order to improve the forecasting of Meta's adjusted closing stock prices. Unlike a standard ARIMA model, SARIMAX allows for the inclusion of external (exogenous) variables that may influence the target variable.

To help capture potential drivers of stock price movements, a set of exogenous variables was selected. These included both features with strong statistical correlation to Meta's stock price and others that, while not directly related, were considered potentially informative. The current set of features includes:

- `technologyChangesPercentage`, `Gold_price`, `Silver_price`, `JPYUSD_close`, `EURUSD_close`,

- `pe`, `ema`, `williams`,

- `META_stock_news_sentiment`, `META_press_release_sentiment`,

- `NVDA_adjClose`, `NVDA_stock_news_sentiment`, `NVDA_press_release_sentiment`

Sentiment variables were converted to numerical values using a predefined mapping (e.g., "`very_positive`" $\rightarrow$ 2, "`neutral`" $\rightarrow$ 0, "`very_negative`" $\rightarrow$ -1). Missing values were handled using forward and backward filling to maintain temporal continuity.

Manual experimentation was conducted to determine a set of exogenous variables that yielded promising initial results. However, additional time and computational resources could be devoted to optimizing feature selection through more systematic methods.

The SARIMAX model was configured with the following parameters:

- Non-seasonal order: $(p = 2,\ d = 1,\ q = 2)$

- Seasonal order: $(P = 1,\ D = 0,\ Q = 1,\ s = 7)$

This configuration means the model learns from the previous two days of price data, accounts for differencing to remove trends, incorporates residual error patterns, and models weekly seasonality. In essence, the model captures both short-term dependencies and recurring weekly patterns in the stock price. Several order combinations were tested, and this configuration produced the most reliable forecasting results.

To forecast future values, the final values of the exogenous variables were held constant over the forecast horizon (15 trading days), assuming no major external changes. This allowed the model to generate forward-looking price predictions based on both recent internal trends and known external influences.

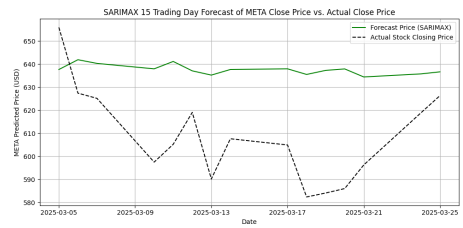

Figure 17: SARIMAX Prediction of closing stock price for META

4.3.5 Model Comparison

Model	Pearson	Spearman	RMSE	MAE	MAPE (%)
Lasso	0.7440	0.4286	391.40	384.09	198.83%
LSTM	0.5095	0.4714	30.46	26.69	4.44%
BiLSTM	0.5127	0.6536	44.26	40.40	6.21%
SARIMAX	0.4418	0.3571	34.88	31.60	4.96%

Table 1: Forecasting Performance Comparison Across Models for Meta

During training, the LSTM, BiLSTM, and SARIMAX models closely approximated the actual stock prices, achieving near-perfect fits with minimal training error. In contrast, the Lasso model exhibited greater difficulty during training, which may help explain its poor forecasting performance relative to the other models.

As seen in Table 1, the BiLSTM model achieved the highest Spearman correlation (0.6536), which serves as the best indicator of the stock movement. Among the models tested, suggesting that it was the most effective at capturing whether the stock price would increase or decrease over the 15-day forecast horizon.

The LSTM model also performed moderately well in this regard, with a Spearman correlation of 0.4714, slightly higher than Lasso (0.4286) and SARIMAX (0.3571). The LSTM model achieved the lowest MAPE (4.44%), indicating that, on average, its forecasts were closest to the actual closing prices in relative terms. Despite the Lasso model having a strong Pearson correlation (0.7440), which reflects the strength of linear association, which is not as relevant for stock data since it typically does not have linear movement. The lasso also had extremely high error values for RMSE (391.4042), MAE (384.0948) and MAPE (198.83%), which is shown in 16 as the forecasted prices were not close to the true values. The results of the LSTM and Bi-LSTM for the Meta data show promise in the ability to predict stock prices and finding the right balance of information and parameters for these models might improve their capabilties.

4.4 GOOGLE

The setup for the Google forecasting model closely followed the approach used for the Meta models. In addition to the lagged features, a set of static features was introduced to evaluate whether their inclusion could enhance forecasting performance. These static features included technologyChangesPercentage, NVDA_adjClose, pe, and Silver_price was moved from the lagged features to the static features. The overall feature set also incorporated lagged stock prices, rolling statistics, momentum indicators, rate of change, sentiment-based features, and engineered interaction terms to capture the relationship between past price movements and sentiment signals.

4.4.1 LSTM

The LSTM model was configured with a single LSTM layer containing 60 units, using one time step and a combination of lagged, rolling, momentum, sentiment, and static features as inputs. The model was compiled with the Adam optimizer and mean squared error (MSE) loss function. To prevent overfitting and preserve optimal weights, early stopping was implemented with a patience of five epochs.

The 15-day forecast was generated recursively, using each predicted value as input for the next step. A weighted update was applied to the first lag value during prediction to simulate more realistic price movement, blending a portion of the previous lag with the newly predicted value.

As shown in the Figure 18, the LSTM model was able to approximate the initial directional movement of Google's stock reasonably well in the early days of the forecast. However, its performance declined in the later stages of the prediction window. Around March 13th, the model began to diverge from the actual stock trajectory, failing to anticipate the upward shift and instead continuing a downward trend. This suggests that while the model captures short-term trends, it struggles to anticipate reversals or turning points in the price series, which is a limitation of simpler unidirectional LSTM architectures when used without enriched future context or exogenous signals. However, trying to add features and layers to the model was also resulting in more erratic results, so more work could go into developing a more optimal LSTM model.

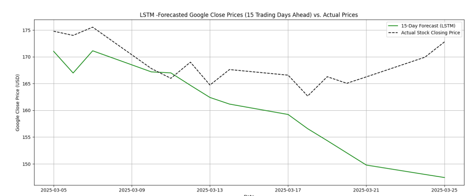

Figure 18: LSTM Prediction of closing stock price for Google

4.4.2 Bi-LSTM

A Bi-LSTM model with a single layer of 50 units was implemented using one time step and compiled with the Adam optimizer and MSE loss. Early stopping was used to prevent overfitting based on validation loss. A mix of lagged, sentiment, rolling, and static features were included in the model input.

The 15-day forecast was generated recursively, with a weighted update applied to the lag structure at each step to improve continuity and realism in predictions. As shown in the graph, the Bi-LSTM captured the initial downward trend but failed to detect the recovery in stock price after mid-March. The forecast increasingly diverged from actual values, suggesting the model struggled with turning points.

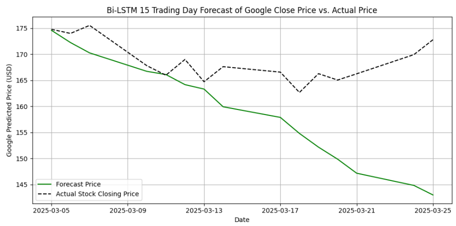

Figure 19: Bi-LSTM prediction of closing stock price for Google

4.4.3 LASSO

The Lasso regression model was trained using cross-validation across the entire dataset to identify the optimal regularization parameter. As shown in the forecast plot, the model performed reasonably well during the 15-day prediction period. However, initial forecasts were overly stable due to the dominance of static features, resulting in a nearly flat forecast line that underestimated the stock price movements.

To address this, small amounts of Gaussian noise were introduced to the static features during forecasting. This adjustment reduced the static features' overpowering influence and allowed the lag features to contribute more meaningfully to short-term variability. While the updated forecasts still struggled to capture inflection points or sharp reversals in price, they better approximated the average trend of the actual closing prices and reflected more realistic day-to-day fluctuations as shown in Figure 20.

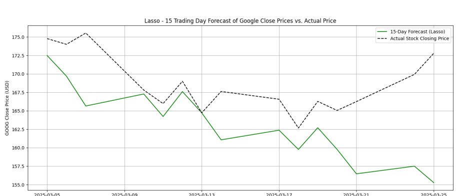

Figure 20: Lasso prediction of closing stock price for Google

4.4.4 SARIMAX

The SARIMAX model for Google's stock was configured with an ARIMA structure of order (2, 1, 2) and a seasonal component (1, 0, 1, 7) to capture weekly patterns. A range of exogenous variables was incorporated, including both features with high correlation or feature importance (identified using a random forest) and others selected for their potential to explain turning points, such as sentiment scores and sector-specific indicators like technologyChangesPercentage.

Initial tests using only highly correlated features resulted in underwhelming performance, likely due to the model's reduced ability to anticipate price reversals. By expanding the feature set to include sentiment and macroeconomic context, the model became better at capturing short-term fluctuations and directional changes.

Figure 21 shows the SARIMAX provided one of the more accurate representations of the actual stock price trend among the models tested, but it still did not capture all the turning points of the graph. While it slightly under-predicted later price increases, it closely followed the general trajectory and captured the initial downturn and mid-period fluctuations with reasonable precision.

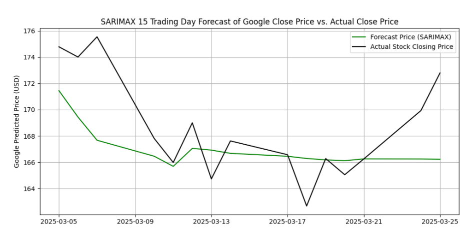

Figure 21: SARIMAX prediction of closing stock price for Google

4.4.5 Model Comparison

Model	Pearson	Spearman	RMSE	MAE	MAPE (%)
Lasso	0.4062	0.3929	7.2714	5.4920	3.43%
LSTM	0.3718	0.3464	11.3926	8.8042	5.21%
BiLSTM	0.3943	0.3821	13.0761	9.4781	5.62%
SARIMAX	0.6828	0.5500	3.4276	2.5102	1.47%

Table 2: Forecasting Performance Comparison Across Models for Google

All models demonstrated strong performance during training, with predicted values closely matching the actual stock prices and low training error across the board. This indicates that each model was able to fit the historical data effectively. However, the forecasting results in Table 2 highlights clear differences in generalization performance. While models like Lasso, LSTM, and BiLSTM performed moderately well, the SARIMAX model's metrics suggest that it was best able to capture the underlying dynamics and short-term fluctuations in Meta's stock price.

The SARIMAX model outperformed the other models across all key metrics. It achieved the highest Pearson (0.6828) and Spearman (0.5500) correlations, indicating strong alignment with the actual price values and directional movement, respectively. Additionally, SARIMAX yielded the lowest RMSE ($3.43), MAE (2.51), and MAPE (1.47%), making it the most accurate model both in terms of magnitude and percentage error.

In contrast, the Lasso model delivered moderate performance, with slightly lower correlation scores but still relatively low RMSE ($7.27) and MAPE (3.43%). Its simplicity may have contributed to greater stability, though it lacked precision in capturing sharper turns or non-linear trends.

The LSTM and BiLSTM models, while designed for capturing sequential patterns, did not outperform the simpler models in this context. Both models had lower correlation values and higher error metrics, particularly RMSE and MAE. This suggests that the LSTM-based models may have struggled with the relatively short forecast horizon or were potentially overfitted during training. Despite their architectural potential, they were less effective than the traditional SARIMAX approach for this specific stock forecasting task. These models show promise and could be useful given the optimal model parameters and features.

4.5 APPLE

Apple's IPO date is from 1980. However, we could not fill some of the news sentiment values and other features even with gpt 4-o model. Hence, to avoid excessive hallucinations from the gpt 4-o and too many missing values, Apple's data started from 1999, 22nd January until the 4th of March, 2025. A time-aware train-test split was done and then the testing set was used to forecast the next 15 trading days following 4th March, 2025. The results of each model are discussed below.

4.5.1 Lasso

Before any training was done with the Lasso regression model, the features were normalised, dummies created and lagged variables were created from a select set of features. The chosen features are based on the economic papers referenced above that these features have a lasting impact on stock prices. They include inflation rates, Gold and Silver prices, Apple news sentiments and the target feature (Apple's closing price).

Similarly to the approach taken for Meta's Lasso regression, we fine-tuned with K-Fold cross-validation to identify the optimal alpha. Once the best alpha (0.0126) was selected, a final model was trained and evaluated using TimeSeriesSplit, a time-aware cross-validation method that preserves the temporal ordering of the data. This ensured the model was validated in a realistic forecasting scenario by only using past data to predict future values.

The 5-fold mean validation root mean squared error (rmse) and testing rmse values are $2.35 and $1.09

respectively. The test data was then used to forecast the next 15 trading days. The figure below compares the forecasted Lasso prices with actual stock prices following 4th March, 2025 over the next 15-trading days.

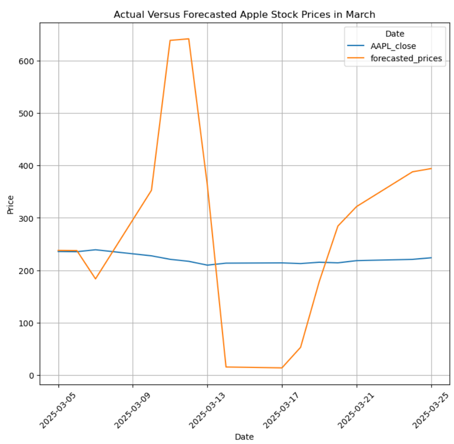

Figure 22: Lasso Prediction of closing stock price for Apple

Lasso performs poorly as it overshoots and underestimates the stock prices. Lasso being a linear model, struggles to capture non-linearities and so tends to perform poorly when forecasting stock prices.

4.5.2 LSTM

Normalization of the features was done as usual, dummies are created and sequences of 90 time steps done as well. Several configurations were tested but the architecture that produced the best results (lowered the test rmse the most and matched patterns between forecast and actual stock prices the most) are as follows:

3 layers of 60, 128 and 32 units respectively all with activations of **tanh** and elastic net kernel regularization of 0.01 each. Dropout rates of 0.3, 0.4 and 0.3 are applied on each layer respectively with Batch normalization applied at each layer after the dropout. Two Dense output layers of units 32 and 1 respectively and activations **relu** and **linear** respectively are used. A learning rate scheduler with initial learning rate of 0.001, decay steps of 10000 and decay rate of 0.9 is used alongside the AdamW optimiser and a batch size of 64 with epochs of 5. The training and test rmses are $42.01 and $52.12 respectively. The figure below compares the forecasted LSTM prices with actual stock prices following 4th March, 2025 over the next 15-trading days.

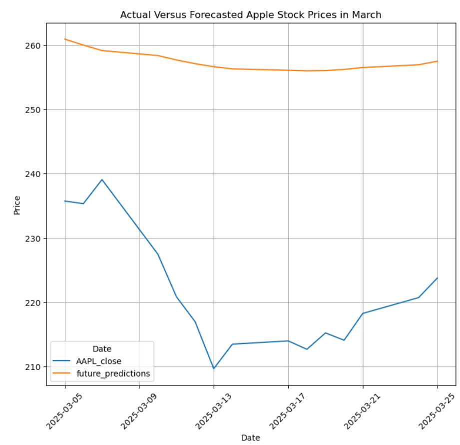

Figure 23: LSTM Prediction of closing stock price for Apple

The LSTM performs great visually as it seems to follow a similar pattern to the actual stock prices for Apple although the dip in the actual stock prices for Apple is bigger than the forecasted prices. However, this visual appeal indicates the LSTM model captures the non-linearities and temporal dependencies in the stock prices significantly well.

4.5.3 Bi-LSTM

The exact process done for LSTM was done for Bi-LSTM. The configurations and/or architecture are the exact same. The only difference was a batch size of 32 and 10 epochs used instead. The training and test rmses are $43.01 and $50.67 respectively. The figure below compares the forecasted Bi-LSTM prices with actual stock prices following 4th March, 2025 over the next 15-trading days.

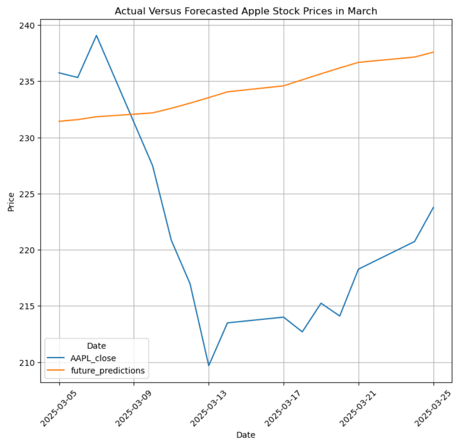

Figure 24: Bi-LSTM Prediction of closing stock price for Apple

The Bi-LSTM performs better than the Lasso model visually but worse than the LSTM model as it fails to capture the dip in the actual stock prices.

Model	Pearson	Spearman	RMSE	MAE	MAPE (%)
Lasso	0.061	0.250	196.008	152.448	69.910
LSTM	**0.925*****	**0.864*****	37.048	36.258	16.556
BiLSTM	-0.554**	-0.454*	**16.585**	**15.093**	**6.951**

Table 3: Forecasting Performance Comparision Across Models For Apple. *, ** and *** indicate 10%, 5% and 1% p-value significance on the correlation metrics respectively.

Table 2 shows that the LSTM model achieved the highest Pearson (0.925) and Spearman correlation (0.6536) all being statistically significant at the 1% level. Thus, among the models tested, suggesting that it was the most effective at capturing whether the stock price would increase or decrease over the 15-day forecast horizon. Despite the LSTM's strong capacity to capture the movements of the stock prices, it was worse off than the Bi-LSTM when being precise about the actual price for the forecasting period as shown with an RMSE of $37.05 compared to the Bi-LSTM with an RMSE of $16.59. That notwithstanding, The rmse for the LSTM is not extreme and given the high correlation and the visual patterns in the graph, it outbeats all the other models tested for Apple.

The lasso model had low correlation values and were insignificant at all levels with high RMSE (196.01), MAE (152.45) and MAPE (69.91%).

4.6 NVIDIA

Nvidia's IPO date is from 1999, 22nd January until the 4th of March, 2025. A time-aware train test split was done and the testing set was used to forecast the next 15 trading days following 4th March, 2025. The results of each model are discussed below.

4.6.1 Lasso

Before any training was done with the Lasso regression model, the features were normalised, dummies created and lagged variables were created from a select set of features. The chosen features are based on the economic papers referenced above that these features have a lasting impact on stock prices. They include inflation rates, Gold and Silver prices, Nvidia news sentiments and the target feature (Nvidia's closing price).

Using a predefined alpha of 0.001 after testing out a couple of others, a model was trained and evaluated using TimeSeriesSplit with three (3) splits ensuring the model was validated in a realistic forecasting scenario by only using past data to predict future values.

The 3-fold mean validation root mean squared error (rmse) and testing rmse values are $1.71 and $4.87 respectively. The test data was then used to forecast the next 15 trading days. The figure below compares the forecasted Lasso prices with actual stock prices following 4th March, 2025 over the next 15-trading days.

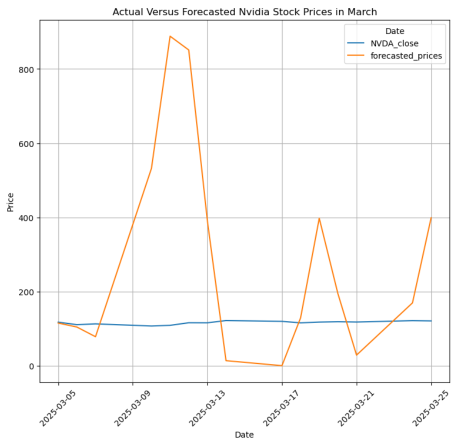

Figure 25: Lasso Prediction of closing stock price for Nvidia

Lasso, similar to the case for other companies, performs poorly as it overshoots and underestimates the stock prices. Lasso being a linear model, struggles to capture non-linearities and so tends to perform poorly when forecasting stock prices.

4.6.2 LSTM

Normalization of the features was done as usual, dummies are created and sequences of 27 time steps done as well. Several configurations were tested but the architecture that produced the best results (lowered the test rmse the most and matched patterns between forecast and actual stock prices the most) are as follows:

2 layers of 30,and 60 units respectively all with activations of 'relu' and elastic net kernel regularization of 200.5 each. Dropout rates of 0.3 and 0.5 are applied on each layer respectively with Batch normalization applied at each layer after the dropout. Two Dense output layers of units 20 and 1 respectively and activations **relu** and **linear** respectively are used. A learning rate scheduler with initial learning rate of 0.00001, decay steps of 10000 and decay rate of 0.9 is used alongside the AdamW optimiser and a batch size of 40 with epochs of 40. The training and test rmses are \$15.02 and \$147.32 respectively. This indicates overfitting but these are much improved results from other configurations tried. With additional time, the results on Nvidia could be improved for the LSTM model. The figure below compares the forecasted LSTM prices with actual stock prices following 4th March, 2025 over the next 15-trading days.

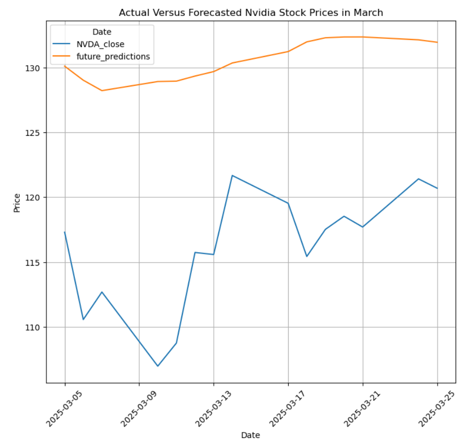

Figure 26: LSTM Prediction of closing stock price for Nvidia

The LSTM performs great visually as it seems to follow a similar pattern to the actual stock prices for Nvidia although the dip in the actual stock prices for Nvidia is bigger than the forecasted prices. However, this visual appeal indicates the LSTM model captures the non-linearities and temporal dependencies in the stock prices significantly well despite the overfitting.

4.6.3 Bi-LSTM

Normalization of the features was done as usual, dummies are created and sequences of 50 time steps done as well. Several configurations were tested but the architecture that produced the best results (lowered the test rmse the most and matched patterns between forecast and actual stock prices the most) are as follows:

2 layers of 30,and 60 units respectively all with activations of relu and elastic net kernel regularization of 200.5 each for the first layer and 300.5 for l1 and

200.5 for l2 in the second layer. Dropout rates of 0.2 and 0.3 are applied on each layer, respectively, with Batch normalization applied at each layer after the dropout. Two Dense output layers of units 30 and 1 respectively and activations **relu** and **linear** respectively are used. A learning rate scheduler with initial learning rate of 0.00001, decay steps of 10000 and decay rate of 0.9 is used alongside the AdamW optimiser and a batch size of 32 with epochs of 25. The training and test rmses are $244.04 and $857.44 respectively. The figure below compares the forecasted Bi-LSTM prices with actual stock prices following 4th March, 2025 over the next 15-trading days.

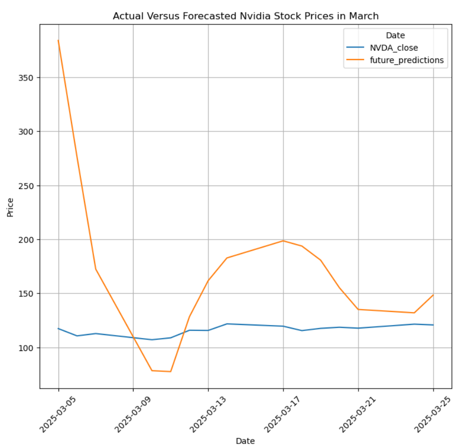

Figure 27: Bi-LSTM Prediction of closing stock price for Nvidia

The Bi-LSTM performs worse than the LSTM model visually as it overestimates and underestimates the actual stock prices.

Model	Pearson	Spearman	RMSE	MAE	MAPE (%)
Lasso	-0.463*	-0.382	327.042	218.069	191.652
LSTM	**0.717***	**0.689***	**14.979**	**14.575**	**11.177**
BiLSTM	0.203	0.154	92.730	65.644	32.731

Table 4: Forecasting Performance Comparision Across Models For Nvidia. *, ** and *** indicate 10%, 5% and 1% p-value significance on the correlation metrics respectively.

Table 2 shows that the LSTM model dominates in all metrics considered and the correlation metrics were significant at the 1% level. The correlation metrics are on average rougly 70% making it a good forecast and again confirming the LSTM model to be a powerful model for forecasting stock prices. The lasso model performed the worst across all metrics as well and the Bi-LSTM model performing better than the lasso model but worse than the LSTM model.

5 Discussion

While no single model outperformed across all stocks and metrics, the LSTM model demonstrated the most consistent performance overall, frequently achieving competitive results in both correlation and error metrics. LSTM's performance stands out due to its ability to capture long-term dependencies and time-series patterns inherent in financial data. For majority of the companies considered, LSTM dominated in most of the metrics considered, thus, making it suitable for time-series forecasting tasks.

This aligns with previous research that highlights LSTM's ability to model complex, nonlinear relationships in time-series data, especially for tasks such as stock price prediction. The performance of LSTM is particularly notable when compared to traditional models like ARIMA and regression-based models such as Lasso. While Lasso regression performed well in terms of correlation metrics, its higher RMSE and MAE indicate that it may struggle to capture the non-linear movement in stock prices, unlike LSTM, which is more adept at modeling sequential dependencies in time-series data.

The poor performance of BiLSTM, ARIMA, and Random Forest models can be attributed to several factors. BiLSTM, though designed to capture both forward and backward dependencies in the data, still shows weak performance in this context. This might be because the additional complexity of BiLSTM did not translate into improved predictions for these specific datasets. ARIMA, while widely used for time-series forecasting, may be less suitable for stock price prediction due to its reliance on assumptions of linearity and stationary conditions that are frequently violated in financial time series. In contrast, the SARIMAX model shows greater promise, as it can incorporate both exogenous variables and seasonal components. This makes it better equipped to capture recurring patterns and external market influences, offering a more flexible and robust framework for modeling stock price behavior compared to traditional ARIMA models. Similarly, Random Forest, while useful in regression tasks, may not fully leverage the sequential nature of stock price movements, which could explain its relatively poor performance compared to LSTM.

The choice of features used in the models, such as past stock prices, moving averages, and technical indicators—may not fully capture all external factors that influence stock price movements, including macroeconomic indicators, geopolitical developments, and broader market sentiment. Additionally, the decision to treat certain features as lagged inputs versus static variables had a measurable impact on forecast accuracy. This highlights the importance of thoughtful feature engineering, as the structure and temporal treatment of input variables directly affect a model's ability to capture both short-term fluctuations and longer-term trends. The varying performance of different models under different preprocessing pipelines, feature selection, and model tuning strategies further emphasizes the critical role of data preparation and model configuration. Continued experimentation with feature selection and transformation is necessary to identify optimal combinations that enhance predictive performance.

Future research could explore the incorporation of additional features macroeconomic indicators, to improve model performance. Furthermore, exploring hybrid models that combine the strengths of different algorithms, such as combining LSTM with ARIMA or Random Forest, may offer improved prediction accuracy. Finally, applying these models to a broader range of industries or global markets could provide insights into the generalizability of the findings and further validate the effectiveness of LSTM in stock price prediction.

Also, work on dealing with the gpt4-o model hallucinations could also be done. This is not a major concern as it was only used for filling in missing values in the news sentiment. However, accounting for any hallucinations may improve the results for all datasets. In conclusion, the results indicate that LSTM is the most effective model for stock price forecasting among the models tested. Its ability to capture sequential patterns in the data makes it well-suited for time-series prediction tasks, although future research may benefit from incorporating additional features and exploring hybrid modeling approaches.

6 Conclusion

The evaluation of various forecasting models demonstrated that different approaches offer advantages depending on the specific stock and prediction objective. While LSTM and BiLSTM models are effective at capturing short-term patterns and directional trends, they often struggle with volatility and turning points unless carefully tuned. The SARIMAX model, which incorporates both autoregressive components and exogenous variables, achieved the most accurate and stable forecasts for Google. BiLSTM performed best for Meta in capturing directional shifts, and LSTM showed strong predictive ability for NVIDIA and Apple. In contrast, Lasso regression performed moderately but was generally outperformed by models better suited to capturing non-linear relationships. The inclusion of news sentiment features added value to the forecasting process, especially when combined with technical indicators and domain-relevant variables. Future work should focus on refining feature engineering—particularly the use of static versus lagged variables—and exploring hybrid models that can blend statistical and deep learning techniques to improve generalizability and robustness.

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